

Configuration of RIP and OSPF using Cisco packet tracer and verify the connectivity

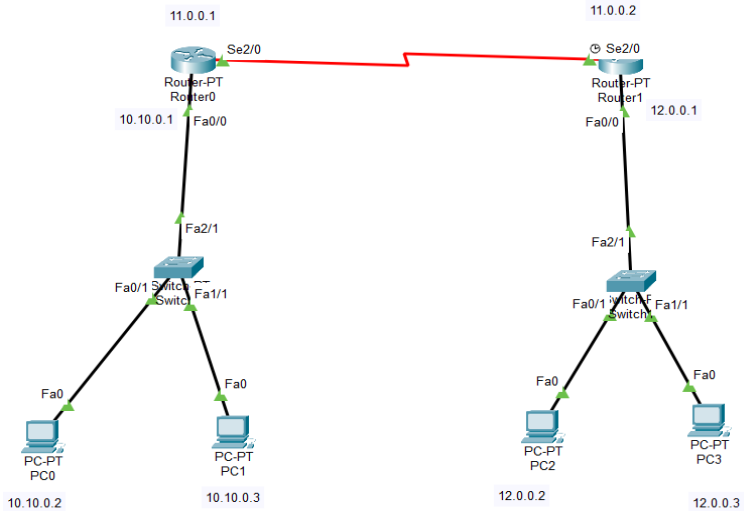


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RIP Configuration





Devices and Connections

- Select two routers (Router-PT) and place them on the workspace
- Select two switches (Switch-PT) and place them on the workspace
- Select four PCs and place them on the workspace
- Connections for left-side switch
 - Use a Copper straight-through and connect
 - FastEthernet of PC0 $\Rightarrow$ FastEthernet0/1 of the switch
 - FastEthernet of PC1 $\Rightarrow$ FastEthernet1/1 of the switch
 - FastEthernet2/1 of switch to FastEthernet0/0 of the left-side router
- Connections for right-side switch
 - Use a Copper straight-through and connect
 - FastEthernet of PC2 $\Rightarrow$ FastEthernet0/1 of the switch
 - FastEthernet of PC3 $\Rightarrow$ FastEthernet1/1 of the switch
 - FastEthernet2/1 of switch to FastEthernet0/0 of the right-side router
- Use "Serial DTE" cable to connect the Serial2/0 (left router) $\leftrightarrow$ Serial2/0 (right router)



Assign IP Address

Click on PC → Desktop Tab → IP Configuration

- PC0: IPv4: 10.10.0.2; SubMask:255.0.0.0; Gateway: 10.10.0.1
- PC1: IPv4: 10.10.0.3; Sub Mask:255.0.0.0; Gateway: 10.10.0.1
- PC2: IPv4: 12.0.0.2; Sub Mask:255.0.0.0; Gateway: 12.0.0.1
- PC3: IPv4: 12.0.0.3; Sub Mask:255.0.0.0; Gateway: 12.0.0.1



Configuring Gateway of router (Left)

Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status: ☐ 100 Mbps ☐ 10 Mbps ☒ On

Bandwidth: ☐ Half Duplex ☐ Full Duplex ☒ Auto

Duplex: ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address: 0060.3E87.BA1A

IP Configuration

IPv4 Address: 10.10.0.1

Subnet Mask: 255.0.0.0

Tx Ring Limit: 10

Equivalent IOS Commands

```
Router(config-if)#no shutdown
Router(config-if)#ip address 10.10.0.1 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 10.10.0.1 255.0.0.0
Router(config-if)#
```



Configuring Serial Interface of router (Left)

Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 11.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

Press RETURN to get started!

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#no shutdown
Router(config-if)#ip address 11.0.0.1 255.0.0.0
Router(config-if)#
```



RIP Configuration of the router (left)

```
Router(config)#router rip
Router(config-router)#network 10.10.0.0
Router(config-router)#network 11.0.0.0
Router(config-router)#do write
Building configuration...
[OK]
Router(config-router)#exit
Router(config)#sh ip rip database
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip rip database
10.0.0.0/8      auto-summary
10.0.0.0/8      directly connected, FastEthernet0/0
11.0.0.0/8      auto-summary
11.0.0.0/8      directly connected, Serial2/0
```




Configuring Gateway of router (Right)

Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.4A42.BD6B

IP Configuration

IPv4 Address 12.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
ip address 11.0.0.2 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 12.0.0.1 255.0.0.0
Router(config-if)#
```



Configuring Serial Interface of router (Right)

Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status

Duplex ☒ Full Duplex ☒ On

Clock Rate 1200

IP Configuration

IPv4 Address 11.0.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
ip address 11.0.0.2 255.0.0.0
Router(config-if)#
```



RIP Configuration of the router (Right)

```
-
Router(config-if)#
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 11.0.0.0
Router(config-router)#network 12.0.0.0
Router(config-router)#do write
Building configuration...
[OK]
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip rip database
10.0.0.0/8      auto-summary
10.0.0.0/8
    [1] via 11.0.0.1, 00:00:21, Serial2/0
11.0.0.0/8      auto-summary
11.0.0.0/8      directly connected, Serial2/0
12.0.0.0/8      auto-summary
12.0.0.0/8      directly connected, FastEthernet0/0
Router#
```



Check Connection

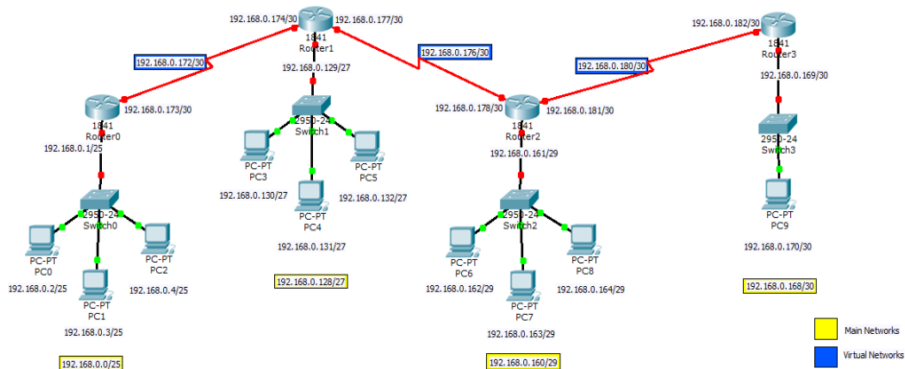
- Ping the PCs to check connection. Every PC is connected to every other PC
- Inspect the Routing table
- "sh ip route rip" in the CLI of the router
- Demo: <https://www.youtube.com/watch?v=C5sM6wCATnA>



Exercise: Implement RIP routing

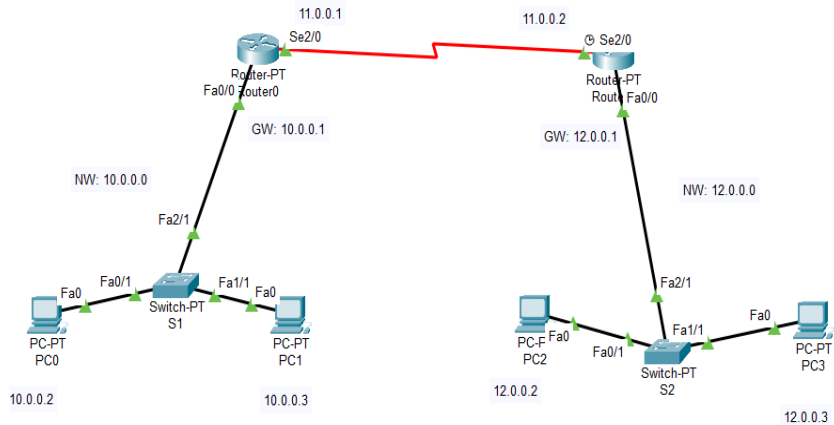
<https://eiheducation.in/>

how-to-configure-rip-routing-on-4-routers-in-cisco-packet-tr





OSPF Configuration





OSPF Areas

An area is a logical grouping of networks, routers, and links that share the same area ID.

Benefits:

- Reduces the number of routes that need to be propagated
- Reduces the size of the topology database that each router needs to maintain
- Simplifies administration
- Optimizes traffic and resource utilization
- <https://www.geeksforgeeks.org/configuring-multi-area-ospf-in-cisco/>

Note: In this experiment, entire network is considered "**area 0**"



OSPF Process ID

An OSPF process ID is a number that identifies different OSPF processes on a router. It's a key part of network engineering

- The OSPF process ID range is 1 to 65535.
 - The syntax for the OSPF process ID is "router ospf process-id".
- Network engineers may run multiple OSPF processes to segment different types of traffic for security or organizational purposes.



Devices and Connections

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 - FastEthernet of PC2 $\Rightarrow$ FastEthernet0/1 of the switch
 - FastEthernet of PC3 $\Rightarrow$ FastEthernet1/1 of the switch
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Assign IP Address

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- PC0: IPv4: 10.10.0.2; SubMask:255.0.0.0; Gateway: 10.10.0.1
- PC1: IPv4: 10.10.0.3; Sub Mask:255.0.0.0; Gateway: 10.10.0.1
- PC2: IPv4: 12.0.0.2; Sub Mask:255.0.0.0; Gateway: 12.0.0.1
- PC3: IPv4: 12.0.0.3; Sub Mask:255.0.0.0; Gateway: 12.0.0.1



Configuring Gateway of router (Left)

Router0

Physical **Config** CLI Attributes

GLOBAL

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- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status: ☐ 100 Mbps ☐ 10 Mbps ☒ On

Bandwidth: ☐ Half Duplex ☐ Full Duplex ☒ Auto

Duplex: ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address: 0060.3E87.BA1A

IP Configuration

IPv4 Address: 10.10.0.1

Subnet Mask: 255.0.0.0

Tx Ring Limit: 10

Equivalent IOS Commands

```
Router(config-if)#no shutdown
Router(config-if)#ip address 10.10.0.1 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 10.10.0.1 255.0.0.0
Router(config-if)#
```



Configuring Serial Interface of router (Left)

Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 11.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

Press RETURN to get started!

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#no shutdown
Router(config-if)#ip address 11.0.0.1 255.0.0.0
Router(config-if)#
```



OSPF Configuration of the router (left)

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 255.0.0.0 area 0
Router(config-router)#network 11.0.0.0 255.0.0.0 area 0
```

Figure: OSPF Configuration

```
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    11.0.0.0/8 is directly connected, Serial2/0
O    12.0.0.0/8 [110/65] via 11.0.0.2, 00:05:17, Serial2/0
```

Figure: Checking the ip route

```
Router#sh ip route ospf
```



Configuring Gateway of router (Right)

Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.4A42.BD6B

IP Configuration

IPv4 Address 12.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
ip address 11.0.0.2 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 12.0.0.1 255.0.0.0
Router(config-if)#
```



Configuring Serial Interface of router (Right)

Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 11.0.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
ip address 11.0.0.2 255.0.0.0
Router(config-if)#
```




OSPF Configuration of the router (Right)

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 11.0.0.0 255.0.0.0 area 0
Router(config-router)#network 12.0.0.0 255.0.0.0 area 0
00:29:43: %OSPF-5-ADJCHG: Process 1, Nbr 11.0.0.1 on Serial2/0 from LOADING to FULL, Loading Done
```

Router> sh ip route ospf

O 10.0.0.0 [110/65] via 11.0.0.1, 00:23:17, Serial2/0



Check Connection

- Ping the PCs to check connection. Every PC is connected to every other PC
- Inspect the Routing table
- "sh ip route ospf" in the CLI of the router
- Demo: <https://www.youtube.com/watch?v=2y2IvslS51E>



Conclusion

we successfully configured RIP and OSPF on Cisco network switches and Routers using Cisco Packet Tracer. Configuring both RIP and OSPF enhances the network's ability to dynamically learn and advertise routes, improving overall network efficiency and reliability.